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1. Always check the cables

```
→ ~ ping 130.208.246.98
PING 130.208.246.98 (130.208.246.98) 56(84) bytes of data.
64 bytes from 130.208.246.98: icmp_seq=1 ttl=52 time=5.30 ms
64 bytes from 130.208.246.98: icmp_seq=2 ttl=52 time=5.58 ms
64 bytes from 130.208.246.98: icmp_seq=3 ttl=52 time=4.61 ms
^C
--- 130.208.246.98 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 4.606/5.160/5.578/0.408 ms
→ ~ sudo nmap -sS 130.208.246.98
Starting Nmap 7.92 ( https://nmap.org ) at 2026-08-25 17:58 GMT
Nmap scan report for 130.208.246.98
Host is up (0.0047s latency).
Not shown: 884 filtered tcp ports (no-response), 114 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
5050/tcp   open  mmcc

Nmap done: 1 IP address (1 host up) scanned in 3.55 seconds
→ ~ □
```

(b)

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5.

```
sudo tcpdump -vAX -i wlp0s20f3 'port 5001'
```

```
→ ~ sudo tcpdump -vAX -i wlp0s20f3 'port 5001'
[sudo] password for birkir:
dropped privs to tcpdump
tcpdump: listening on wlp0s20f3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
19:03:13.622473 IP (tos 0x0, ttl 64, id 38336, offset 0, flags [DF], proto TCP (6), length 60)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [S], cksum 0x7e4f (incorrect -> 0xa4
c7), seq 3105210140, win 64240, options [mss 1460,sackOK,TS val 998425072 ecr 0,nop,wscale 7],
length 0
    0x0000:  4500 003c 95c0 4000 4006 26db c0a8 4445  E..<..@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1c 0000 0000  ...b.....
    0x0020:  a002 faf0 7e4f 0000 0204 05b4 0402 080a  ....~0.....
    0x0030:  3b82 c1f0 0000 0000 0103 0307                ;.....
19:03:13.627579 IP (tos 0x0, ttl 52, id 0, offset 0, flags [DF], proto TCP (6), length 60)
    130.208.246.98.complex-link > mini-mekka.41208: Flags [S.], cksum 0x627c (correct), seq 14
73383690, ack 3105210141, win 65160, options [mss 1460,sackOK,TS val 3165658389 ecr 998425072,n
op,wscale 7], length 0
    0x0000:  4500 003c 0000 4000 3406 c89b 82d0 f662  E..<..@.4.....b
    0x0010:  c0a8 4445 1389 a0f8 57d2 0d0a b915 bf1d  ..DE....W.....
    0x0020:  a012 fe88 627c 0000 0204 05b4 0402 080a  ....b|.....
    0x0030:  bcb0 1d15 3b82 c1f0 0103 0307                ....;.....
19:03:13.627608 IP (tos 0x0, ttl 64, id 38337, offset 0, flags [DF], proto TCP (6), length 52)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [.], cksum 0x7e47 (incorrect -> 0x8d
d5), ack 1, win 502, options [nop,nop,TS val 998425078 ecr 3165658389], length 0
    0x0000:  4500 0034 95c1 4000 4006 26e2 c0a8 4445  E..4..@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1d 57d2 0d0b  ...b.....W...
    0x0020:  8010 01f6 7e47 0000 0101 080a 3b82 c1f6  ....~G.....;...
    0x0030:  bcb0 1d15                ....
19:04:35.553365 IP (tos 0x0, ttl 64, id 38338, offset 0, flags [DF], proto TCP (6), length 95)
    mini-mekka.41208 > 130.208.246.98.complex-link: Flags [P.], cksum 0x7e72 (incorrect -> 0xc
0ee), seq 1:44, ack 1, win 502, options [nop,nop,TS val 998507003 ecr 3165658389], length 43
    0x0000:  4500 005f 95c2 4000 4006 26b6 c0a8 4445  E..._@.&...DE
    0x0010:  82d0 f662 a0f8 1389 b915 bf1d 57d2 0d0b  ...b.....W...
    0x0020:  8018 01f6 7e72 0000 0101 080a 3b84 01fb  ....~r.....;...
    0x0030:  bcb0 1d15 5265 6d65 6d62 6572 2074 6f20  ....Remember.to.
    0x0040:  6368 6563 6b20 666f 7220 6e65 7477 6f72  check.for.networ
    0x0050:  6b20 636f 6e6e 6563 7469 7669 7479 0a    k.connectivity.
19:04:35.557912 IP (tos 0x0, ttl 52, id 6331, offset 0, flags [DF], proto TCP (6), length 52)
    130.208.246.98.complex-link > mini-mekka.41208: Flags [.], cksum 0x0d92 (correct), ack 44,
win 509, options [nop,nop,TS val 3165740319 ecr 998507003], length 0
    0x0000:  4500 0034 18bb 4000 3406 afe8 82d0 f662  E..4..@.4.....b
    0x0010:  c0a8 4445 1389 a0f8 57d2 0d0b b915 bf48  ..DE....W.....H
    0x0020:  8010 01fd 0d92 0000 0101 080a bcb1 5d1f  .....].
    0x0030:  3b84 01fb                ;...
```

(c) In your own words, explain what each of the three commands does. Hint: it may help to read the manual (man page) for ncat and tcpdump.

```
sudo tcpdump -vAX -i ens192 'port 5000':
```

Here we have a tcpdump command used to capture and display network traffic on an interface. ran with sudo since capturing raw traffic needs elevated privileges. It is paired with the Options -v for slightly more verbose output. -A which prints each packet's data in ASCII and -X which

additionally prints it in hex. -i ens192 tells it which interface to listen on and 'port 5000' filters it down to only traffic on that port.

```
ncat -ln 5000:
```

Here we have a ncat command which alone is a command used to read and write data across networks but here it is also paired with the Options -l which makes it listen for connections instead of connecting to a remote machine which is the default and -n which disables hostname resolution so it skips resolving names to IPs and vice versa rather than requiring one.

```
ncat 130.208.246.98 5000
```

Here the ncat command without any Options so it has the Hostname which was disabled in the last command and the port aswell which is the default way of using Ncat, so this particular command connects to whatever is listening on that IP and port and lets you exchange data with it interactively

(e) How does the tcpdump output differ when you used UDP vs TCP? Why?

The main difference is that UDP sends the data in a single packet without establishing a connection, while TCP establishes a connection and uses acknowledgements.

- **UDP:** One packet is sent directly:
 - proto UDP (17)
 - Contains the 43-byte message immediately.
 - No handshake or ACK packets.
- **TCP:** Several packets are visible:
 - SYN -> starts the connection.
 - SYN-ACK -> server accepts it.
 - ACK -> completes the TCP three-way handshake.
 - PSH-ACK -> sends the n-byte message.
 - ACK -> confirms the data was received.

Output

UDP:

```
19:32:50 IP mini-mekka.37567 > 130.208.246.98: UDP, length 43  
"Remember to check for network connectivity"
```

TCP:

```
19:33:11 IP mini-mekka.52038 > 130.208.246.98: Flags [S] # SYN
```

```
19:33:11 IP 130.208.246.98 > mini-mekka.52038: Flags [S.] # SYN-ACK
19:33:11 IP mini-mekka.52038 > 130.208.246.98: Flags [.] # ACK
19:33:29 IP mini-mekka.52038 > 130.208.246.98: Flags [P.], length 43
    "Remember to check for network connectivity"
19:33:29 IP 130.208.246.98 > mini-mekka.52038: Flags [.], ack 44 # ACK
```